

Chapter 2 – Identity Management

Learning Objectives

After completing this chapter, you should be able to:

- Explain how identity works in Google Workspace.
- Design Organizational Units (OUs).
- Design Groups.
- Explain the differences between OUs and Groups.
- Understand Admin Roles.
- Explain Delegated Administration.
- Design enterprise identity architecture.
- Avoid common design mistakes.

1. Introduction

Identity is the heart of Google Workspace.

Every action performed inside Google Workspace begins with an identity.

Examples

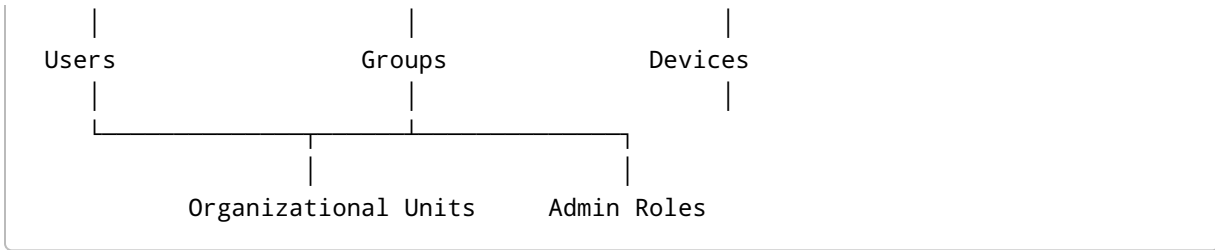
- Sending email
- Accessing Drive
- Joining Meet
- Sharing files
- Using APIs
- Signing into third-party applications

Everything starts with:

User Identity

Identity Architecture





Everything revolves around identity.

2. Users

A user represents a person or service identity inside the tenant.

Example

rahul@company.com

A user receives

- Gmail
- Drive
- Calendar
- Meet
- Chat
- Contacts
- Licenses
- Policies

Users authenticate to access Workspace services.

User Lifecycle

Every enterprise should define a lifecycle.

HR Request

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Create User

↓

Assign License

↓

Assign OU

↓

Assign Groups

↓

User Starts Work

↓

Role Changes

↓

Department Changes

↓

Termination

↓

Suspend

↓

Archive/Migrate Data

↓

Delete User

Enterprise Example

A new employee joins the Finance department.

The administrator:

Creates:

john@company.com

Assigns

- Enterprise Plus License

Moves user to

Finance OU

Adds Groups

- finance@company.com
- payroll@company.com

Applies

- MFA
- Context-Aware Access
- DLP

Everything is automatically applied based on the identity.

3. Organizational Units (OUs)

Organizational Units are used to apply policies.

Think of an OU as a policy container.

Example

Company

- └─ HR
- └─ Finance
- └─ Engineering
- └─ Sales
- └─ Contractors

Policies can differ between OUs.

Example

Engineering

- Allow GitHub
- Allow Developer Extensions

HR

- Restrict external sharing
- Strong DLP

Contractors

- Limited Drive Sharing
 - No Admin Console Access
 - Restricted API Access
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Important Rule

A user can belong to

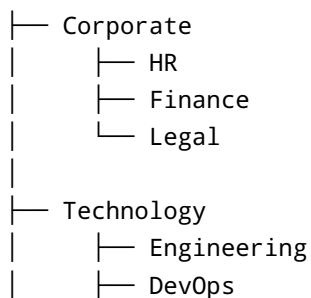
ONLY ONE

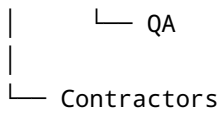
Organizational Unit.

This is one of the biggest differences between OUs and Groups.

OU Hierarchy

Company





Policies inherit from parent OUs unless overridden.

4. Groups

Groups are designed for collaboration and permissions.

Unlike OUs,

Users may belong to multiple Groups.

Example

Rahul

belongs to

```
engineering@company.com  
developers@company.com  
security@company.com  
project-alpha@company.com
```

This is perfectly normal.

Why Groups Exist

Suppose

Engineering

contains

2,000 employees.

Only 20 work on Project Alpha.

Should you create another OU?

No.

Create

project-alpha@company.com

This is exactly what Groups are for.

Group Types

Common examples

All Employees

HR

Engineering

Finance

Developers

Security

Project Teams

Support Teams

Groups are also used for

- Email distribution
 - Permission management
 - Drive access
 - Calendar sharing
 - IAM assignments
 - Application access
-

Enterprise Example

Finance Team

finance@company.com

Permissions

- Shared Drive
- Shared Calendar
- Shared Mailbox

When a new employee joins,
add them to the Group.

Permissions are inherited automatically.

5. OUs vs Groups

One of the most common interview questions.

Organizational Unit	Group
Used for policies	Used for collaboration
One user belongs to one OU	User may belong to many Groups
Hierarchical	Flat membership
Security settings	Resource access
Device policies	Email distribution
License targeting (where applicable through admin processes)	File and Calendar sharing

Example

Engineering OU

Contains

500 Developers

Policies

- MFA
- Chrome Settings
- Context-Aware Access

Groups

developers@

backend@

frontend@

devops@

security@

Policies come from the OU.

Permissions come from Groups.

6. Admin Roles

Not everyone should be a Super Admin.

Google provides delegated admin roles.

Examples

Super Admin

User Management Admin

Groups Admin

Help Desk Admin

Services Admin

Why?

Suppose HR needs to reset passwords.

Do they need Super Admin?

No.

Assign

Help Desk Admin.

Least privilege.

Admin Role Architecture

Super Admin

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├─ User Admin

├─ Groups Admin

├─ Help Desk

├─ Security Admin

└─ Reports Admin

Every administrator receives only the permissions required.

7. Delegated Administration

Large organizations cannot rely on one administrator.

Example

Company

50,000 users

Offices

India

USA

Germany

Australia

Each regional IT team receives delegated administration.

India Admin

Can manage

India OU

Cannot modify

USA OU

This improves security and operational efficiency.

8. Secondary Domains

One tenant may manage multiple domains.

Example

Primary

company.com

Secondary

company.in

company.co.uk

company.au

All users belong to one Workspace tenant.

9. Domain Alias

Suppose the company owns

company.com

company.org

Users should receive mail on both.

Instead of creating separate users,

configure

company.org

as a Domain Alias.

Now

john@company.com

automatically receives mail sent to

john@company.org

Secondary Domain vs Domain Alias

Secondary Domain	Domain Alias
Independent user accounts	Alternate email address

Secondary Domain	Domain Alias
Different usernames possible	Same users receive both addresses
Used after mergers or acquisitions	Used for branding or alternate domains
Separate identities	Alias for existing identities

10. Enterprise Identity Design

Small Company

Users

↓

Single OU

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Simple Groups

Enterprise

Tenant

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├─ OUs

|

├─ HR

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├─ Finance

|

├─ Engineering

|

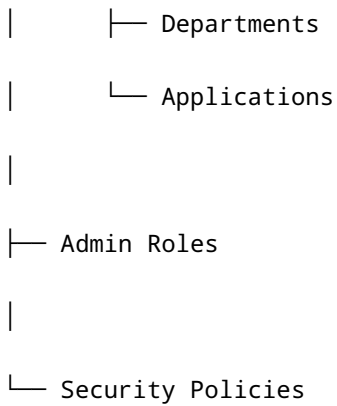
└─ Contractors

|

├─ Groups

|

├─ Projects



A diagram showing the components of an identity architecture. It consists of a vertical line on the left with several horizontal lines branching off to the right. The branches are labeled from top to bottom: 'Departments', 'Applications', 'Admin Roles', and 'Security Policies'. There is an additional unlabeled branch between 'Applications' and 'Admin Roles'.

- Departments
- Applications
- Admin Roles
- Security Policies

Best Practices

- ✓ Keep the OU hierarchy simple.
- ✓ Use Groups for permissions.
- ✓ Use OUs for policies.
- ✓ Avoid unnecessary nesting.
- ✓ Create custom admin roles.
- ✓ Review admin privileges regularly.
- ✓ Design for future growth.

Common Mistakes

- ✗ Using OUs instead of Groups.
- ✗ Making every admin a Super Admin.
- ✗ Deep OU hierarchies.
- ✗ Creating hundreds of unnecessary Groups.
- ✗ Not documenting the identity architecture.

Real Enterprise Scenario

A company acquires another organization with 3,000 users.

Steps

1. Assess both tenants.
2. Decide whether to add the acquired domain as a Secondary Domain or Domain Alias.
3. Plan OU structure.
4. Create Groups.
5. Configure SSO and SCIM.
6. Pilot migration.
7. Migrate in phases.
8. Validate permissions and policies.

Identity planning comes before migration.

Troubleshooting

When a user cannot access a service, ask:

1. Does the user exist?
 2. Is the correct license assigned?
 3. Is the user in the correct OU?
 4. Is the user in the correct Group?
 5. Does the user have the required admin role?
 6. Are security policies blocking access?
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Interview Questions

1. What is the difference between an OU and a Group?
2. Why can a user belong to only one OU?
3. Why shouldn't everyone receive Super Admin privileges?

4. When would you use a Secondary Domain instead of a Domain Alias?
 5. How would you design identity for a company with 50,000 users?
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Chapter Summary

In this chapter you learned

- ✓ Identity Architecture
- ✓ Users
- ✓ Organizational Units
- ✓ Groups
- ✓ Admin Roles
- ✓ Delegated Administration
- ✓ Secondary Domains
- ✓ Domain Aliases
- ✓ Enterprise Identity Design

The next chapter will explore **Authentication and Authorization**, where you'll learn how Google verifies identity, grants access, and integrates with external Identity Providers using SAML, OAuth, and OpenID Connect.